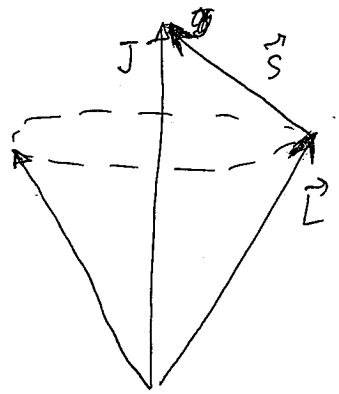
Weak magnetic field (Zeeman effect)

$$H_Z = -(\vec{\mu}_e + \vec{\mu}_s)\cdot\vec{B} = +\frac{e}{2m}(\vec{L}+2\vec{S})\cdot\vec{B}$$

$$\vec{\mu}_s = -\frac{ge}{2m}\vec{S}, \quad \vec{\mu}_e = -\frac{e}{2m}\vec{L}$$

① If $H_{so} = \lambda\,\vec{S}\cdot\vec{L} \gg H_Z \Rightarrow \vec{L}, \vec{S}$ are not separately conserved but $\vec{J} = \vec{L}+\vec{S}$ is!



$$\frac{d}{dt}\vec{L} = \frac{i}{\hbar}[H,\vec{L}]$$

$$[H_{so},\vec{L}] = \lambda[L^a S^a, L^b] =$$

$$= \lambda[L^a, L^b]S^a = \lambda\hbar i\,\varepsilon^{abc}S^a L^c$$

$$= -i\lambda\hbar(\vec{S}\times\vec{L})$$

$$\Rightarrow [H_{so},\vec{S}] = -i\lambda\hbar(\vec{L}\times\vec{S})$$

$$[H_{so}, \underbrace{\vec{L}+\vec{S}}_{\vec{J}}] = -i\lambda\hbar(\vec{S}\times\vec{L}+\vec{L}\times\vec{S}) = 0$$

$$\vec{L}+2\vec{S} = \vec{J}+\vec{S}$$

We need projection of $\vec{S}$ on $\vec{J}$ (projection $\vec{S}\perp\vec{J}$ rotates around J-axis and averages to zero)

let $\left.\begin{aligned}\vec{S} &= a\vec{J}+\vec{S}_\perp\\ \vec{L} &= b\vec{J}+\vec{L}_\perp\end{aligned}\right\}$ $\vec{S}+\vec{L} = \vec{J} = (a+b)\vec{J}+\vec{S}_\perp+\vec{L}_\perp$

$\Rightarrow a+b=1$

$\vec{S}_\perp = -\vec{L}_\perp$

Also $a = \dfrac{\vec{J}\cdot\vec{S}}{(\vec{J})^2}$

$$\Rightarrow \langle\vec{S}\rangle = \left\langle\frac{\vec{J}\cdot\vec{S}}{J^2}\vec{J}\right\rangle \Rightarrow$$

$$\vec{L} = \vec{J} - \vec{S} \quad \Rightarrow \quad \vec{L}^2 = \vec{J}^2 + \vec{S}^2 - 2\vec{J}\vec{S}$$

$$\Rightarrow \quad \vec{J}\cdot\vec{S} = \frac{\hbar^2}{2}\Big(J(J+1) + S(S+1) - L(L+1)\Big)$$

$$H_z = \frac{e}{2m}\left(\vec{J} + \frac{\vec{J}\cdot\vec{S}}{\vec{J}^2}\,\vec{J}\right)\cdot\vec{B} =$$

$$= \frac{e}{2m}\left(1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2\,j(j+1)}\right)\vec{J}\cdot\vec{B}$$

$$\Rightarrow \langle H_z\rangle = \underbrace{\left(\frac{e\hbar}{2m}\right)}_{\mu_B} g_j\, m_j\, B$$

where $\hbar m_j = J_z$

$$g_j = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)} \qquad \text{Lande g-factor}$$

$$= \frac{3}{2} + \frac{S(S+1) - L(L+1)}{J(J+1)}$$

(2) Strong field so that $H_z \gg H_{so}$

$$H_z = \frac{e}{2m}\left(\vec{L} + 2\vec{S}\right)\cdot\vec{B} \quad \text{and} \quad H_{so} = \lambda\,\vec{L}\cdot\vec{S} \text{ is small perturbation}$$

$$\Rightarrow \langle H_z\rangle = \frac{e\hbar}{2m}\left(m_l + 2m_s\right)B = \frac{e\hbar}{2m}\left(L_z + 2S_z\right)B$$

and $\langle H_{so}\rangle = \lambda\,\langle\vec{L}\rangle\cdot\langle\vec{S}\rangle = \lambda\hbar^2\, L_z\cdot S_z$

Slightly different way, using Wigner-Eckart Th:

$$H_Z = \frac{e}{2m}\left(\vec{L} + g\vec{S}\right)\cdot\vec{B} = \frac{\mu_B}{\hbar}\left(L_z + gS_z\right)B$$

Choosing $\vec{B} = B\hat{z}$. Note that $\vec{L} + g\vec{S} \neq \vec{J} = \vec{L} + \vec{S}$

(1) $H_{so} = \lambda\vec{S}\cdot\vec{L} \gg H_Z \Rightarrow$ need to evaluate matrix elements of perturbation H_Z in the basis that diagonalizes $H + H_{so}$:

$$\langle j, m; l, s|\ (L_z + gS_z)\ |j, m'; l, s\rangle \qquad m \equiv j_z$$

Wigner-Eckart Theorem (all vectors are alike)

Matrix element of any vector in a $(2j+1)$-dimensional subspace of eigenstates $|j, m\rangle$ are proportional to matrix element of $\vec{J}$ itself!

(*) $$\langle j, m; l, s|\left(L_z + gS_z\right)|j, m'; l, s\rangle = g(jls)\,\langle j, m; l, s|\,J_z\,|j, m'; l, s\rangle$$

the proportionality constant does not depend on m, m' (does not depend on orientation of the coordinate system)

Calculate $g(j, l, s)$ with the help of Wigner-Eckart:

Multiply both sides by $\langle j, m'; l, s|\vec{J}|j, m; l, s\rangle$: and sum over m'

$$\sum_{m'} \langle j, m; l, s|\left(\vec{L} + g\vec{S}\right)|j, m'; l, s\rangle\langle j, m'; l, s|\,\vec{J}\,|j, m; l, s\rangle =$$

$$= \sum_{m'} g(jls)\,\langle j, m; l, s|\,\vec{J}\,|j, m'; l, s\rangle\langle j, m'; l, s|\,\vec{J}\,|j, m; l, s\rangle$$

Then right-hand side can be written as

$$\text{rhs} = g(j\ell s) \sum_{m'} \langle j,m;\ell,s|\vec{J}|j,m';\ell,s\rangle\langle j,m';\ell,s|\vec{J}|j,m;\ell,s\rangle$$

But we know that $\langle j,m;\ell,s|\vec{J}|j',m';\ell,s\rangle = 0$ for $j \neq j'$

$\Rightarrow$ extend the sum $\sum_{m'} \rightarrow \sum_{jm'}$.

Now note that $\sum_{jm'} |j,m';\ell,s\rangle\langle j,m';\ell,s| = 1$

because we sum over all $(2\ell+1)(2s+1)$ states!

The same summation is of course done on the <u>left hand side</u>

$$\Rightarrow \langle j,m;\ell,s|(\vec{L}+g\vec{S})\cdot\vec{J}|j,m;\ell,s\rangle = g(j\ell s)\langle j,m;\ell,s|\vec{J}\cdot\vec{J}|j,m;\ell,s\rangle$$

Next $\vec{S}^2 = (\vec{J}-\vec{L})^2 = \vec{J}^2 + \vec{L}^2 - 2\vec{L}\cdot\vec{J}$

$$\vec{L}^2 = (\vec{J}-\vec{S})^2 = \vec{J}^2 + \vec{S}^2 - 2\vec{S}\cdot\vec{J}$$

$$\Rightarrow \vec{L}\cdot\vec{J} = \frac{1}{2}\left(\vec{J}^2 + \vec{L}^2 - \vec{S}^2\right)$$

$$\vec{S}\cdot\vec{J} = \frac{1}{2}\left(\vec{J}^2 + \vec{S}^2 - \vec{L}^2\right)$$

$$\Rightarrow \frac{1}{2}\langle j,m;\ell,s|\left[\vec{J}^2 + \vec{L}^2 - \vec{S}^2 + g\left(\vec{J}^2 + \vec{S}^2 - \vec{L}^2\right)\right]|j,m;\ell,s\rangle$$

$$= \frac{1}{2}\left[(1+g)\,j(j+1) + (1-g)\,\ell(\ell+1) - (1-g)\,s(s+1)\right] = g(j\ell s)\,j(j+1)$$

$$\Rightarrow g(j,\ell,s) = \frac{1}{2}\left[(1+g) + (1-g)\,\frac{\ell(\ell+1) - s(s+1)}{j(j+1)}\right]$$

For $g=2$ $\quad g(j,\ell,s) = \frac{3}{2} + \frac{s(s+1) - \ell(\ell+1)}{j(j+1)}$ $\quad$ Laude g-factor